# Deep Learning for Skin Cancer Image Analysis: A Systematic Literature Review

## Methodology

**Literature Search**

*PRISMA 2020 guideline*
*Scopus, IEEE, ACM, Springer, ArXiv*

**Screening**

*Quantitative Impact Filter*
*DL Eligibility Assessment*

**Included Studies**

*n = 234 Primary Studies*

## DL Diagnostic Pipeline

**Data Foundation**

*ISIC Archive, HAM10000,*
*PH2, PAD-UFES-20*

**Preprocessing & Augmentation**

*Resizing, Artifact Removal & Geometric Augmentation*

**Architectural Shift**

*Foundational CNNs →*
*Vision Transformers & Hybrid Models*

**Computational Tasks**

**Segmentation:** *Lesion & Attributes*
**Classification:** *Binary & Multi-class*

## Clinical Translation

**Persistent Challenges**

*- Severe Data Imbalance*
*- Poor OOD Generalization*
*- Opaque AI (Black-Box Models)*
*- High Computational Overhead*

**Future Directions**

*- Explainable AI (XAI)*
*- Multimodal Patient Data Integration*
*- Federated Learning (FL)*
*- Real-World Clinical Validation*